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Project : Biological databases and Sequence alignment

Exercise 1 - Retrieve hemoglobin α and β sequences from UniProt

Hemoglobin is a tetramer composed of two α chains and two β chains, each binding a heme group required for oxygen transport.

Database/tool. UniProt Knowledgebase (UniProtKB). We preferentially selected **reviewed (Swiss-Prot)** entries because they are expert-curated and generally provide higher reliability and richer cross-references than automatically annotated records (**Exercise 1, Figure 4 in Appendix).**

Procedure.

1. Queried UniProt for "hemoglobin subunit alpha Homo sapiens" and opened the reviewed entry.
2. Recorded the UniProt accession, entry name, and sequence length displayed in the entry header.
3. Downloaded the protein sequence in FASTA format.
4. Repeated the same process for "hemoglobin subunit beta Homo sapiens".

Results.

Table 1 Uniprot Report on Hemoglobin Subunits

Subunit	UniProt accession	Entry name	Organism	UniProt status	Length (aa)
Hemoglobin α	P69905	HBA_HUMAN	Homo sapiens	Reviewed (Swiss-Prot)	**142**
Hemoglobin β	P68871	HBB_HUMAN	Homo sapiens	Reviewed (Swiss-Prot)	**147**

Critical Note. Reporting **accessions** (e.g., P69905, P68871) is essential because names like "hemoglobin alpha" can refer to multiple related genes/isoforms; accessions uniquely identify the exact database objects used, making the analysis reproducible

Exercise 2 - Protein–protein interactions of hemoglobin

Table 2 Number of protein interactors reported for hemoglobin subunits across databases

Protein subunit	UniProt accession	Database	Interactors reported	Notes
Hemoglobin β	P68871	BioGRID	~226–228	Includes high- and low-throughput physical interactions
Hemoglobin β	P68871	IntAct	~120	Includes curated interactions and complex expansion
Hemoglobin α (HBA1)	P69905	BioGRID	~75	Gene-specific interaction records
Hemoglobin α (HBA2)	P69905	BioGRID	~143	Separate gene product, same protein sequence
Hemoglobin α	P69905	IntAct	>100	Aggregated protein-level interaction network

NB : Complete database query screenshots are provided in Exercise 2, Figure 5 of Appendix

To investigate the protein-protein interaction landscape of hemoglobin, the human hemoglobin α (**UniProt: P69905**) and β (**UniProt: P68871**) subunits were queried across multiple interaction databases. Screenshots presented (in Appendix), show the interaction summaries obtained from **UniProt cross-references**, **BioGRID**, and **IntAct**. The aim was to determine how many interactors are reported for each subunit and to understand why these numbers differ between databases.

For the **hemoglobin β subunit (HBB, P68871)**, UniProt cross-references indicate many reported interactors, with **BioGRID listing approximately 226-228 interacting proteins**, while **IntAct reports around 120 interaction partners**. Direct inspection of the BioGRID results reveals that these interactions are supported by **76 publications** and are dominated by **high-throughput physical interaction evidence**, with additional low-throughput and chemical interaction data. In contrast, the **IntAct** interaction network presents a smaller but still dense set of interaction partners, visualized as a network where edges represent curated interaction evidence rather than raw publication counts.

For the **hemoglobin α subunit (P69905)**, the situation is more complex. BioGRID separates the interactions of this UniProt entry into two gene products, **HBA1 and HBA2**, which encode highly

similar α-globin proteins. As a result, BioGRID reports **approximately 75 unique interactors for HBA1** and **143 unique interactors for HBA2**, with different numbers of raw interaction events and publications supporting each. IntAct similarly displays an extensive interaction network for hemoglobin α, incorporating direct interactions as well as complex-based associations. UniProt itself does not provide a single definitive interactor count for hemoglobin α because it aggregates interaction evidence from multiple external databases and gene products.

The observed differences in interactor numbers across databases do not reflect biological contradictions but instead arise from **fundamental differences in database design and curation strategies**. BioGRID is a **literature-driven resource** that aims to capture all **experimentally reported interactions**, including results from **large-scale high-throughput studies**. Consequently, it often reports higher numbers of interactors, as each published interaction contributes to the total count. IntAct, by contrast, applies more structured curation rules and focuses **on standardized molecular interaction records**, which typically results in **lower but more curated interactor counts**.

An additional source of discrepancy lies in **how interactions are defined and counted**. Some databases count **unique interacting partners**, while others count **interaction events**, meaning that the same protein pair supported by multiple experiments or publications may be counted multiple times. Furthermore, databases such as IntAct frequently perform **complex expansion**, whereby proteins that are part of the same molecular complex are listed as interacting with one another, even if no direct physical binding experiment was performed for each pair.

In the specific case of hemoglobin α, the existence of two highly similar genes (HBA1 and HBA2) further contributes to variability in interaction counts. Although these genes encode nearly identical protein sequences, they are treated separately in gene-centric databases, leading to distinct interaction records that are only partially merged at the protein level. This illustrates a key principle of biological databases: **gene-centric and protein-centric representations can yield different quantitative results**, even when describing the same biological entity.

Overall, the large number of reported interactors for hemoglobin subunits should be interpreted with caution. Hemoglobin is a well-studied, abundant protein that participates in stable complexes and is frequently detected in high-throughput experiments. Therefore, many reported interactions likely reflect **co-complex association, experimental co-purification, or indirect evidence**, rather than direct and specific physical binding. This exercise highlights why interaction numbers are inherently database-dependent and why careful interpretation is essential when comparing results across different biological databases.

Exercise 3 - Biological context of BRCA1 and 53BP1 (UniProt)

The biological roles of BRCA1 and 53BP1 were examined using their reviewed UniProt entries, with emphasis on their function in the DNA damage response (**available screenshots of the query are found in Exercise 3, Figure 6 of the Appendix**). Both proteins are involved in the repair of DNA double-strand breaks (DSBs), but they regulate **distinct and opposing repair pathways**.

BRCA1 is a nuclear protein with **E3 ubiquitin–protein ligase activity**, which is regulated by phosphorylation. UniProt annotations indicate that BRCA1 plays a central role in **homologous recombination (HR)**, a high-fidelity DNA repair pathway that ensures accurate repair of DSBs. By promoting DNA end resection and coordinated repair, BRCA1 contributes to genome stability, explaining why its dysfunction is strongly linked to cancer predisposition.

53BP1 (TP53BP1) is also recruited to DNA double-strand breaks but promotes the alternative **non-homologous end joining (NHEJ)** pathway. UniProt shows that 53BP1 recognizes specific histone modifications at damaged chromatin and, upon phosphorylation by ATM, accumulates at DSB sites. Importantly, 53BP1 counteracts BRCA1-mediated homologous recombination by limiting DNA end resection, thereby favoring faster but more error-prone repair.

In summary, UniProt annotations reveal that BRCA1 and 53BP1 function within the same DNA repair process but act as **functional antagonists**. The balance between BRCA1-driven HR and 53BP1-driven NHEJ determines how cells respond to DNA damage, with major implications for genome stability and cancer development.

Exercise 4 - Dotplot analysis of BRCA1 and 53BP1 using EMBOSS dotmatcher

Dotplots were generated to compare the protein sequences of BRCA1 and 53BP1 using the EMBOSS dotmatcher tool, with different values of **wordsize** and **threshold**. The resulting plots (shown above) illustrate how parameter choices influence the detection and visualization of sequence similarity **(Exercise 4, Figure 7, Appendix).**

When a **large wordsize and high threshold** were used (wordsize = 20, threshold = 50), the dotplot is almost empty, showing only a very small number of isolated dots. This indicates that BRCA1 and 53BP1 do not share long stretches of highly similar amino acid sequences. At these stringent settings, only strong and extended similarities are detected, and since the two proteins are not close homologs, most matches are filtered out.

Using **intermediate parameter values** (wordsize = 10, threshold = 30) produces a dotplot with a few scattered dots and short diagonal fragments. These weak and localized signals suggest the

presence of short regions with limited similarity, but no global alignment or extended conserved domains. The background remains mostly clear, indicating a balance between sensitivity and specificity.

At **small wordsize and low threshold** (wordsize = 7, threshold = 20), the dotplot becomes dense and noisy, with many scattered dots across the entire plot. This reflects a high sensitivity to short sequence matches, many of which arise by chance, especially in large proteins. While this setting can reveal short motifs or low-complexity similarities, it also introduces substantial background noise, making biological interpretation more difficult.

Overall, the **wordsize** controls the length scale of similarity detection: smaller wordsizes increase sensitivity to short matches but also increase random noise, whereas larger wordsizes require longer conserved regions and therefore produce cleaner plots. The **threshold** controls stringency: lower thresholds allow weak matches to appear, while higher thresholds restrict the plot to only the strongest similarities. Together, these dotplots show that BRCA1 and 53BP1 share no strong global sequence similarity, consistent with their distinct protein architectures, despite their involvement in the same biological pathway.

Exercise 5 - Global alignment of BRCA1 and 53BP1 using EMBOSS Needle

In a global alignment, **gap penalties** determine how the algorithm balances mismatches and insertions or deletions. The **gap opening penalty** controls how difficult it is to introduce a new gap: high values result in fewer gaps and force more mismatches, while low values allow gaps to be introduced easily. The **gap extension penalty** controls how long a gap can become, with high values favoring short gaps and low values allowing long gap stretches. Together, these parameters strongly influence the structure and score of the alignment.

A global sequence alignment between BRCA1 and 53BP1 was performed using EMBOSS Needle to assess their overall sequence similarity and to evaluate the effect of gap penalties on the alignment **(Exercise 5, Figure 8, Appendix).** As expected from the dotplot analysis, the alignment reveals **low overall sequence similarity**, with many gaps and short local matches distributed across the alignment rather than a continuously conserved region. This confirms that BRCA1 and 53BP1 are not globally homologous proteins, despite their involvement in the same biological pathway.

Changing the **gap opening and gap extension penalties** has a pronounced effect on the alignment structure. When gap penalties are lower, gaps are introduced more readily, resulting in alignments with many insertions and deletions that allow short local similarities to be aligned. This can slightly improve apparent alignment continuity but at the cost of introducing numerous gaps, reducing

biological interpretability. In contrast, higher gap penalties discourage gap formation, leading to alignments with fewer gaps but more mismatches, as the algorithm is forced to align non-matching residues rather than opening new gaps **(Exercise 5, Figure 8, Appendix)**.

Across different parameter settings, the **percentage identity remains low**, and the alignment score varies substantially depending on the balance between mismatches and gaps. Importantly, altering gap penalties does not reveal strong global similarity between BRCA1 and 53BP1, but instead highlights how global alignment algorithms attempt to force alignment over entire sequence lengths even when true homology is weak or absent. These results are consistent with the dotplot analysis and reinforce the conclusion that BRCA1 and 53BP1 share functional relatedness rather than extensive sequence conservation.

Exercise 6 - Understanding dotplots of hemoglobin and "secret" sequences (from first principles)

A **dotplot** is a simple visual tool used to compare two biological sequences. One sequence is placed on the horizontal axis and the other on the vertical axis. Whenever a stretch of amino acids in one sequence matches a stretch in the other, a dot is drawn. If two sequences are **identical**, these matching dots line up to form a **single straight diagonal line** from the bottom-left to the top-right of the plot. Any interruption, break, or extra diagonal indicates a **difference** between the sequences, such as a mutation, insertion, or deletion (**Exercise 6, Figure 8, Appendix).**

Hemoglobin α vs secret α: what the plot is telling us

When comparing the normal hemoglobin α sequence with the secret α sequence, the dotplots show a **perfect, uninterrupted diagonal line**, regardless of whether a small or large wordsize and threshold are used. This visual pattern means that **every part of the normal α sequence matches the corresponding part of the secret α sequence in the same order**. There are no breaks, shifts, or secondary diagonals (**Exercise 6, Figure 8, Appendix)**.

Scientifically, this is strong evidence that the two sequences are **identical or nearly identical at the amino-acid level**. If even a single amino acid were changed, the dotplot would show a small disruption or weakening of the diagonal at that position, especially when using sensitive parameters. Since no such disruption is observed, the most reasonable conclusion is that **the secret α chain does not contain a mutation** relative to the normal hemoglobin α chain.

Hemoglobin β vs secret β: why the plot looks slightly different

For the hemoglobin β comparison, the dotplots again show a **strong diagonal**, indicating that the two sequences are **almost identical overall**. This is expected, because hemoglobin β proteins are highly conserved. However, when stricter parameters are used (larger wordsize and higher threshold), the diagonal is no longer perfectly uniform. Instead, **small additional diagonal fragments or slight disruptions appear** (**Exercise 6, Figure 8, Appendix).**

This happens because dotplots do not compare single amino acids; they compare **short blocks of amino acids ("words")**. If **one amino acid changes**, every word that overlaps that position is affected. At low stringency, the surrounding similarity hides this effect, but at higher stringency, the mismatch becomes visible as a local disturbance in the diagonal.

From a biological standpoint, this pattern is consistent with a **single point mutation**: one amino acid in the secret β sequence differs from the normal hemoglobin β sequence, while the rest of the protein remains unchanged.

Exercise 7 - Comparison of dotplot and global alignment approaches

Using global alignment, the comparison between the normal hemoglobin α subunit and the secret α subunit shows **no difference at all**, with 100% identity across the entire sequence. This confirms that the α chains are identical and contain no mutation. In contrast, the global alignment between the normal hemoglobin β subunit and the secret β subunit reveals **a single amino-acid difference**, with 146 identical residues out of 147 (99.3% identity). The alignment clearly shows a substitution of glutamic acid (E) by valine (V) at position 6 of the β chain, while the rest of the sequence is unchanged (**Exercise 7, Figure 9, Appendix).**

The difference is **much easier to detect using the global alignment** than with the dotplot. While the dotplot is useful for visually detecting overall similarity and roughly localizing a mutation, the disruption caused by a single amino-acid change is subtle and depends strongly on parameter settings. In contrast, the global alignment provides a **direct, residue-by-residue comparison**, making the exact position and nature of the mutation immediately obvious. Therefore, dotplots are more useful for exploratory, qualitative analysis, whereas global alignments are more precise and informative for identifying specific sequence differences.

Exercise 8 - Interpretation of the mutation using BLOSUM62 and physico-chemical properties

The global alignment identified a single substitution in the hemoglobin β chain, where **glutamic acid (E)** is replaced by **valine (V)**. In the BLOSUM62 substitution matrix, the E to V substitution has a **negative score (−2)**, indicating that this change occurs less frequently than expected by chance in conserved protein regions and is therefore considered **non-conservative**. This is consistent with the very different physico-chemical properties of the two residues: glutamic acid is a **polar, negatively charged, hydrophilic** amino acid that often participates in electrostatic interactions, whereas valine is **non-polar, uncharged, and hydrophobic**, typically contributing to hydrophobic packing. The substitution therefore removes a negative charge and replaces it with a hydrophobic side chain, representing a substantial chemical change despite the otherwise high overall sequence similarity.

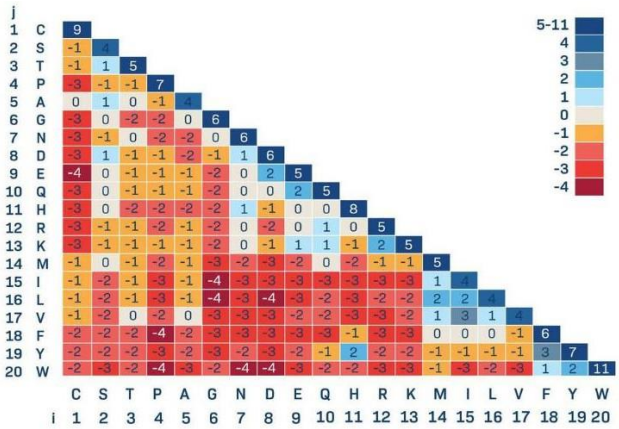


*Figure 1Blossum62 Matrix illustrating a **negative score (−2) for the E - V substitution***

Exercise 9 - Identification of the secret protein using BLAST

Because the global alignment showed that the α subunit is identical to its secret counterpart, BLAST analysis was performed only on the **secret β-chain sequence**, which contains the amino-acid difference. The BLASTP search was run against the **human (taxid:9605)** protein database, ensuring that only human proteins were considered. This avoids cross-species confusion and allows direct identification of the secret protein variant.

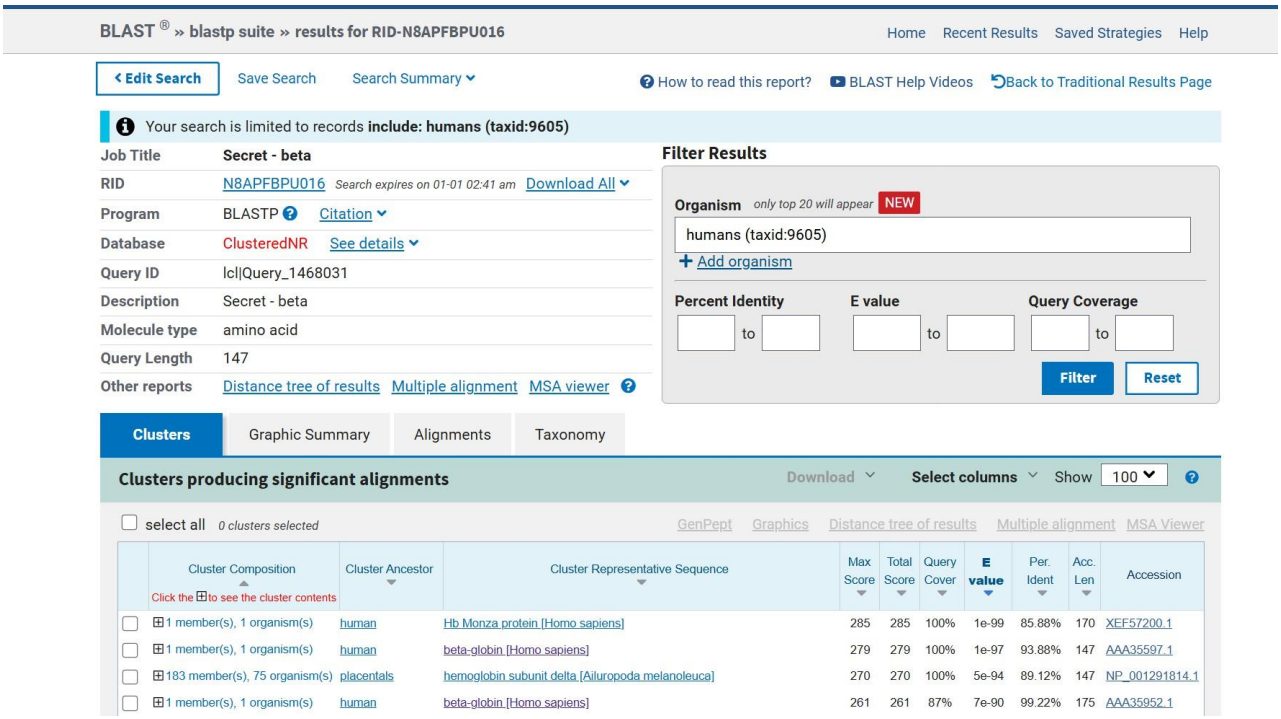


*Figure 2 Blast screenshot showing top 4 searches corresponding to **hemoglobin beta-related proteins**, with extremely E-values and high query coverage, highlighting significant matches*

The BLAST results show that all top matches correspond to **hemoglobin beta-related proteins**, with extremely low E-values and high query coverage, indicating highly significant matches. The **top hit** is annotated as **Hb Monza protein (Homo sapiens)**, showing 100% query coverage and a very strong alignment score. The second hit is annotated as **beta-globin (Homo sapiens)** entries, again with nearly complete query coverage and very high sequence identity. These results clearly indicate that the secret sequence belongs to the **hemoglobin β family**, rather than representing a different protein.

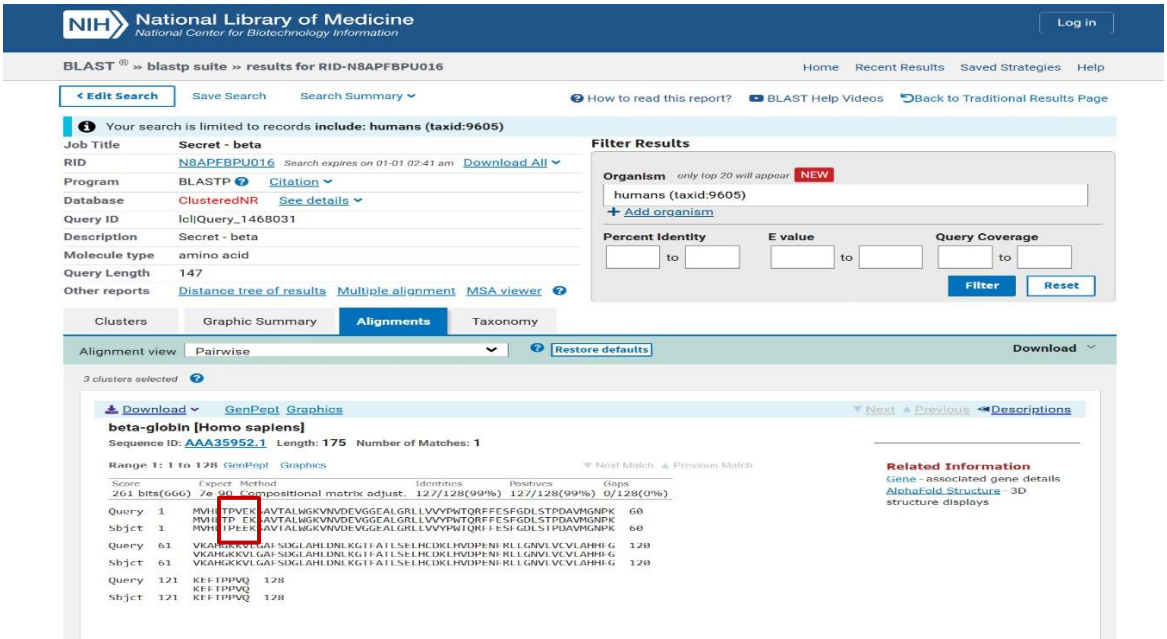


Figure 3 Blast results of the homo sapiens beta-globulin an observed mutation of E to V on position 6

Importantly, although the secret sequence differs from the canonical beta-globin sequence by only a single amino acid, BLAST still correctly identifies it as a beta-globin variant. This is expected, because BLAST prioritizes **overall sequence similarity**, and a single substitution does not change the identity of the protein. The presence of slightly different annotations among the top hits reflects known **hemoglobin variants** rather than distinct protein types. Therefore, the BLAST analysis confirms that the secret protein is a **variant of human hemoglobin subunit beta**, consistent with the mutation identified in previous exercises.

Exercise 10 - Structural comparison and clarification of the mutation

How we arrived at the structures

After identifying the *secret protein* in earlier exercises as a **hemoglobin β-chain variant**, we moved to the **Protein Data Bank (PDB)** to compare its three-dimensional structure with normal hemoglobin. Following the exercise instructions, we first examined **PDB ID: 4HHB**, which corresponds to **human deoxyhemoglobin A**, the normal (wild-type) adult hemoglobin. This structure serves as the reference for canonical hemoglobin architecture and residue composition **(Exercise 10, Figure 12,13, Appendix)**.

Next, based on the identity revealed by BLAST (a hemoglobin β variant), we searched the PDB for structures corresponding to **hemoglobin S (HbS)**, the clinically relevant mutant form. This search led to **PDB ID: 2HBS**, described as *"The high-resolution crystal structure of deoxyhemoglobin S"*. Importantly, this step was not arbitrary: HbS is the best-characterized and most structurally resolved hemoglobin mutation, making it the appropriate structural comparator.

What was observed in the PDB structures (key factual finding)

When inspecting **both 4HHB and 2HBS**, and specifically examining **residue 6 of the β chain**, we observed that **(Exercise 10, Figure 12,13, Appendix)**:

- In **4HHB (normal hemoglobin A)**, residue **β6 is Glutamate (Glu, E)**
- In **2HBS (hemoglobin S)**, residue **β6 is Valine (Val, V)**

This is a **single amino-acid substitution at position β6 (E6V)**.

Why this mutation matters structurally

Glutamate is **negatively charged and hydrophilic**, while valine is **hydrophobic and nonpolar**. Replacing Glu with Val at β6 creates a **hydrophobic patch on the surface of hemoglobin S**, which is absent in normal hemoglobin A. Structural studies have shown that this hydrophobic patch promotes **abnormal intermolecular interactions between hemoglobin molecules in the deoxygenated state**, leading to polymer formation, a hallmark of **sickle-cell disease.**

This mechanistic explanation is **directly supported by crystallographic data**, including the 2HBS structure, and is one of the most well-established structure–function relationships in molecular biology [1, 2]

Overall, I believe this project went beyond the simple application of bioinformatics tools and became an exercise in **scientific reasoning, critical evaluation, and methodological integration**. Rather than treating databases and algorithms as black boxes, we learned to question how results are generated, why different tools produce different outputs, and what each result truly represents biologically. By moving step by step from curated databases to exploratory dotplots, precise alignments, substitution matrices, homology searches, and finally three-dimensional structures, we progressively refined our understanding of hemoglobin at multiple biological scales.

REFERENCES

1. **Suhail, M. (2024). *Biophysical chemistry behind sickle cell anemia and the mechanism of voxelotor action*. *Scientific Reports*, 14, Article 52476. https://www.nature.com/articles/s41598-024-52476-8.pd**
2. **Donkor, A. K., Pagare, P. P., Mughram, M. H. A., & Safo, M. K. (2023). *X-ray crystallography and sickle cell disease drug discovery—a tribute to Donald Abraham*. *Frontiers in Molecular Biosciences*, 10, 1136970. https://doi.org/10.3389/fmolb.2023.1136970**

APPENDIX

Exercise 1

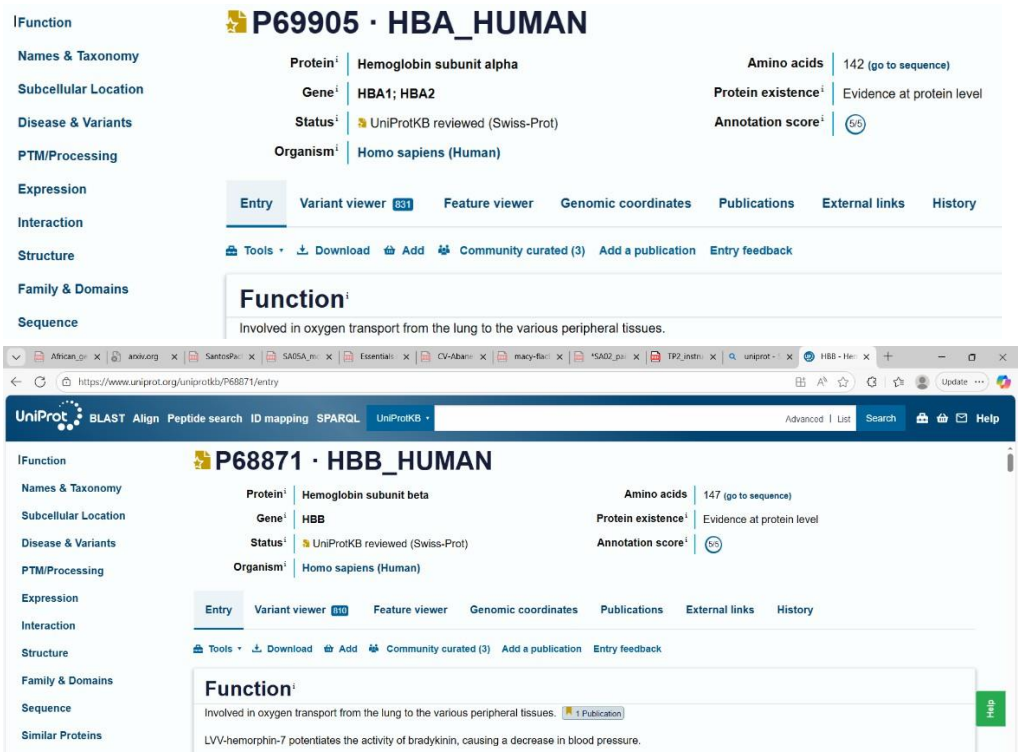
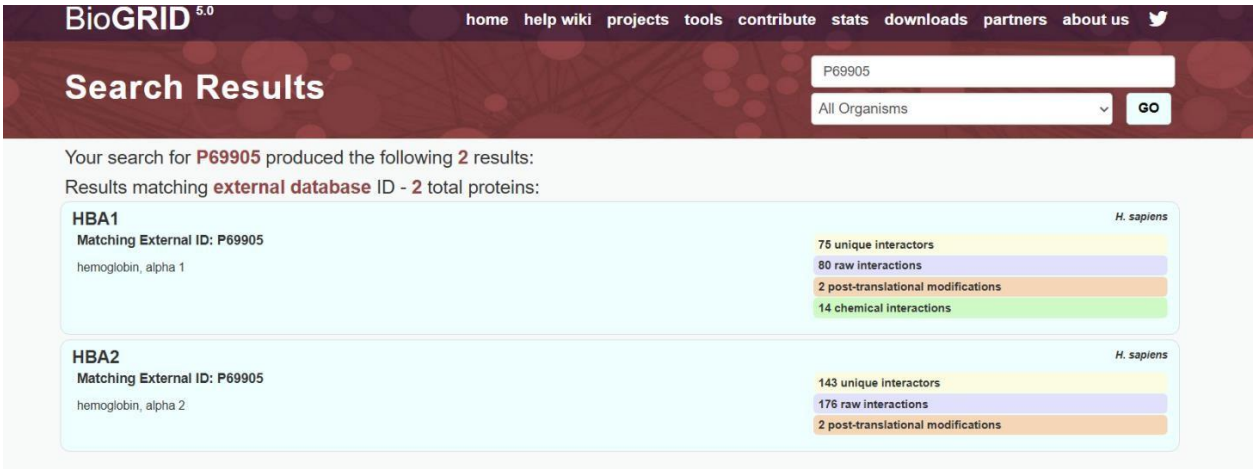
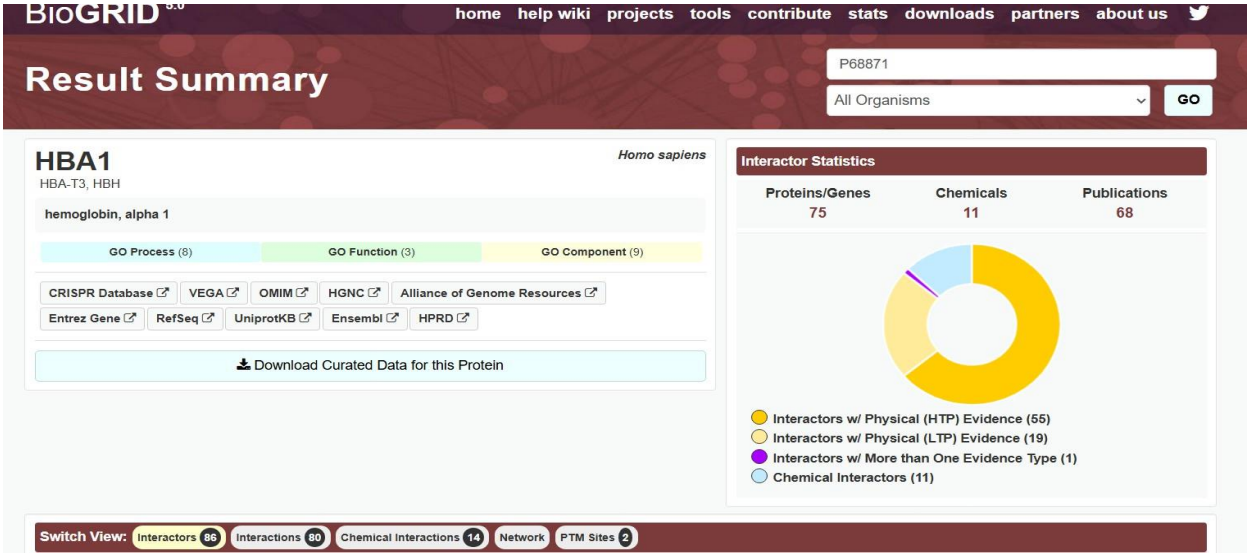


Figure 4 Uniprot Findings of Hemoglobin subunits. Above shows HBA whereas below shows HBB

Exercise 2

Figure 5 Protein–protein interaction networks for hemoglobin α and β subunits retrieved from BioGRID and IntAct databases. Differences in network size reflect database-specific curation strategies, experimental evidence inclusion, and confidence thresholds





BioGRID 5.0

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Result Summary

P68871

All Organisms GO

HBA2

Homo sapiens

HBA-T2, HBH

hemoglobin, alpha 2

GO Process (8) | GO Function (3) | GO Component (9)

CRISPR Database | VEGA | OMIM | HGNC | Alliance of Genome Resources | Entrez Gene | RefSeq | UniprotKB | Ensembl | HPRD

Download Curated Data for this Protein

Interactor Statistics

Proteins/Genes	Publications
143	33

- Interactors w/ Physical (HTP) Evidence (132)
- Interactors w/ Physical (LTP) Evidence (11)

Switch View: Interactors 143 | Interactions 176 | Network | PTM Sites 2

BioGRID 5.0

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Result Summary

P68871

All Organisms GO

HBB

Homo sapiens

CD113t-C, beta-globin

hemoglobin, beta

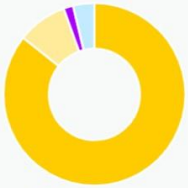
GO Process (13) | GO Function (6) | GO Component (7)

CRISPR Database | OMIM | HGNC | Alliance of Genome Resources | VEGA | Entrez Gene | RefSeq | UniprotKB | Ensembl | HPRD

Download Curated Data for this Protein

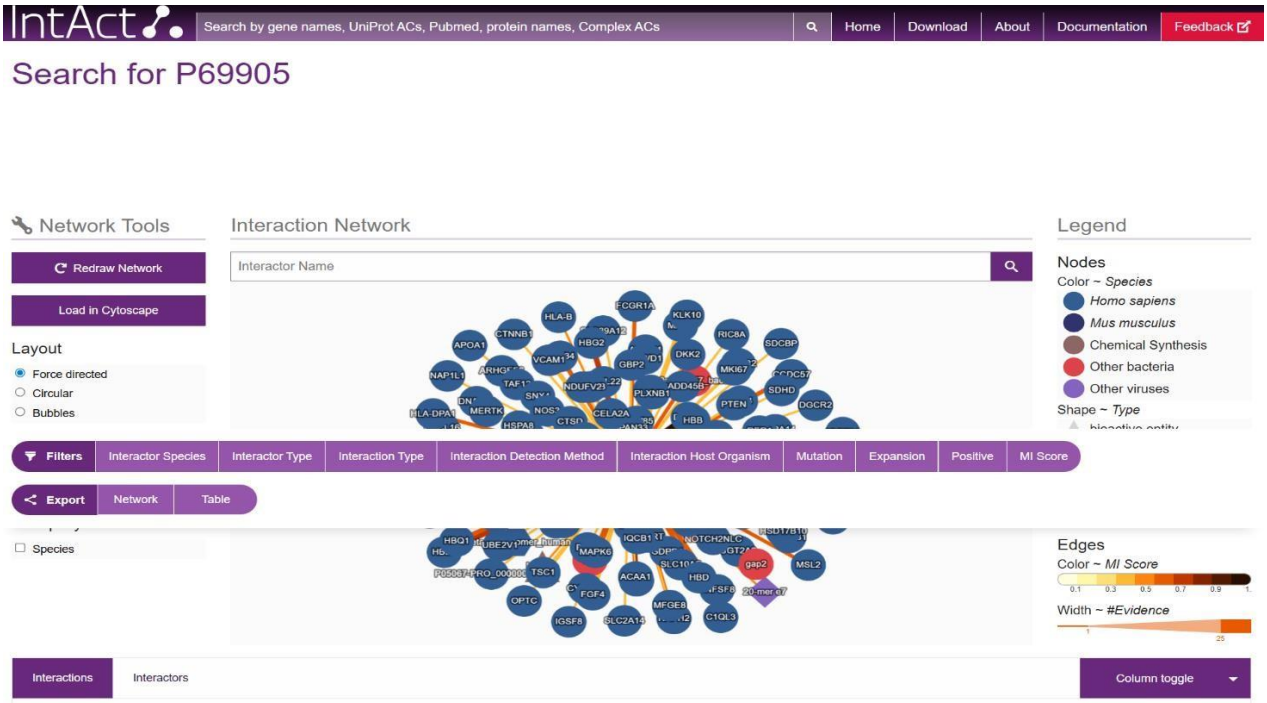
Interactor Statistics

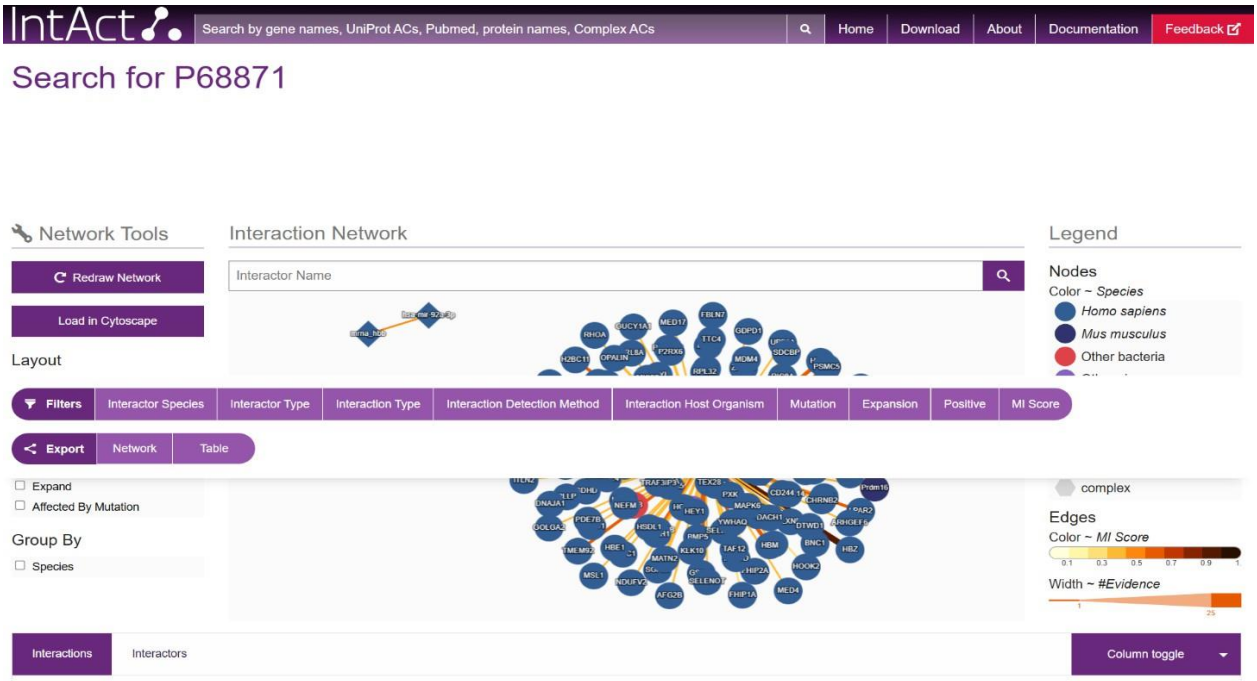
Proteins/Genes	Chemicals	Publications
228	9	76



- Interactors w/ Physical (HTP) Evidence (203)
- Interactors w/ Physical (LTP) Evidence (21)
- Interactors w/ More than One Evidence Type (4)
- Chemical Interactors (9)

Switch View: Interactors 237 | Interactions 279 | Chemical Interactions 10 | Network | PTM Sites 8


IntAct
Search by gene names, UniProt ACs, Pubmed, protein names, Complex ACs
Home
Download
About
Documentation
Feedback
Search for P69905
Network Tools
Redraw Network
Load in Cytoscape
Layout
Force directed
Circular
Bubbles
Interaction Network
Interactor Name
Legend
Nodes
Color ~ Species
Homo sapiens
Mus musculus
Chemical Synthesis
Other bacteria
Other viruses
Shape ~ Type
Filters
Interactor Species
Interactor Type
Interaction Type
Interaction Detection Method
Interaction Host Organism
Mutation
Expansion
Positive
MI Score
Export
Network
Table
Species
Edges
Color ~ MI Score
Width ~ #Evidence
Interactions
Interactors
Column toggle


IntAct
Search by gene names, UniProt ACs, Pubmed, protein names, Complex ACs
Home
Download
About
Documentation
Feedback
Search for P68871
Network Tools
Redraw Network
Load in Cytoscape
Layout
Interaction Network
Interactor Name
Legend
Nodes
Color ~ Species
Homo sapiens
Mus musculus
Other bacteria
Filters
Interactor Species
Interactor Type
Interaction Type
Interaction Detection Method
Interaction Host Organism
Mutation
Expansion
Positive
MI Score
Export
Network
Table
Expand
Affected By Mutation
Group By
Species
complex
Edges
Color ~ MI Score
Width ~ #Evidence
Interactions
Interactors
Column toggle

Exercise 3

Figure c Screenshots of BRCA1(up) and TP53BP1(down) on Uniprot

UniProt BLAST Align Peptide search ID mapping SPARQL UniProtKB Advanced | List Search

Function
Names & Taxonomy
Subcellular Location
Disease & Variants
PTM/Processing
Expression
Interaction
Structure
Family & Domains
Sequence & Isoforms
Similar Proteins

P38398 · BRCA1_HUMAN

Protein: Breast cancer type 1 susceptibility protein
Gene: BRCA1
Status: UniProtKB reviewed (Swiss-Prot)
Organism: Homo sapiens (Human)
Amino acids: 1863 (go to sequence)
Protein existence: Evidence at protein level
Annotation score: 5/5

Entry | Variant viewer 9,944 | Feature viewer | Genomic coordinates | Publications | External links | History

Tools | Download | Add | Community curated (1) | Add a publication | Entry feedback

Function

E3 ubiquitin-protein ligase that specifically mediates the formation of 'Lys-6'-linked polyubiquitin chains and plays a central role in DNA repair by facilitating cellular responses to DNA damage (PubMed:10500182, PubMed:12887909, PubMed:12890688, PubMed:14976165, PubMed:16818604, PubMed:17525340, PubMed:19261748).
It is unclear whether it also mediates the formation of other types of polyubiquitin chains (PubMed:12890688).
The BRCA1-BARD1 heterodimer coordinates a diverse range of cellular pathways such as DNA damage repair, ubiquitination and transcriptional regulation to maintain genomic stability (PubMed:12890688, PubMed:14976165, PubMed:20351172).
Regulates centrosomal microtubule nucleation (PubMed:18056443).
Required for appropriate cell cycle arrests after ionizing irradiation in both the S-phase and the G2 phase of the cell cycle (PubMed:10724175, PubMed:11836499, PubMed:12183412, PubMed:19261748).
Required for FANCD2 targeting to sites of DNA damage (PubMed:12887909).
Inhibits lipid synthesis by binding to inactive phosphorylated ACACA and preventing its dephosphorylation (PubMed:16326698).

UniProt BLAST Align Peptide search ID mapping SPARQL UniProtKB Advanced | List Search

Function
Names & Taxonomy
Subcellular Location
Disease & Variants
PTM/Processing
Expression
Interaction
Structure
Family & Domains
Sequence & Isoforms
Similar Proteins

Gene: TP53BP1
Status: UniProtKB reviewed (Swiss-Prot)
Organism: Homo sapiens (Human)
Protein existence: Evidence at protein level
Annotation score: 5/5

Entry | Variant viewer 2,348 | Feature viewer | Genomic coordinates | Publications | External links | History

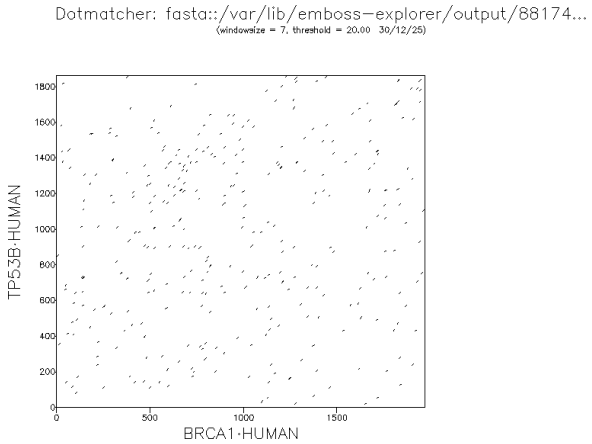
Tools | Download | Add | Add a publication | Entry feedback

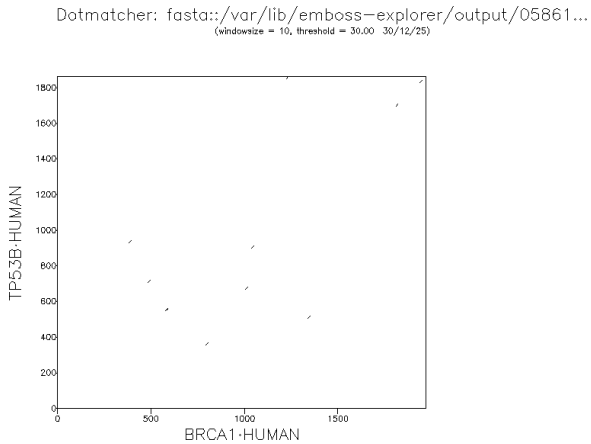
Function

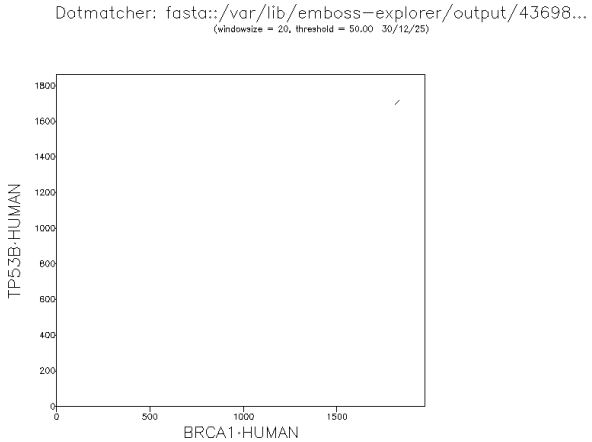
Double-strand break (DSB) repair protein involved in response to DNA damage, telomere dynamics and class-switch recombination (CSR) during antibody genesis (PubMed:12364621, PubMed:17190600, PubMed:21144835, PubMed:22553214, PubMed:23333306, PubMed:27153538, PubMed:28241136, PubMed:31135337, PubMed:37696958).
Plays a key role in the repair of double-strand DNA breaks (DSBs) in response to DNA damage by promoting non-homologous end joining (NHEJ)-mediated repair of DSBs and specifically counteracting the function of the homologous recombination (HR) repair protein BRCA1 (PubMed:22553214, PubMed:23333306, PubMed:23727112, PubMed:27153538, PubMed:31135337).
In response to DSBs, phosphorylation by ATM promotes interaction with RIF1 and dissociation from NUDT16L1/TIRR, leading to recruitment to DSBs sites (PubMed:28241136).
Recruited to DSBs sites by recognizing and binding histone H2A monoubiquitinated at 'Lys-15' (H2AK15Ub) and histone H4 dimethylated at 'Lys-20' (H4K20me2), two histone marks that are present at DSBs sites (PubMed:17190600, PubMed:23760478, PubMed:27153538, PubMed:28241136).
Required for immunoglobulin class-switch recombination (CSR) during antibody genesis, a process that involves the generation of DNA DSBs (PubMed:23345425).
Participates in the repair and the orientation of the broken DNA ends during CSR (By similarity).
In contrast, it is not required for classic NHEJ and V(D)J recombination (By similarity).
Promotes NHEJ of dysfunctional telomeres via interaction with PAXIP1 (PubMed:23727112). By Similarity | 12 Publications

Exercise 4

*Figure 7 BRCA1 and 53BP1 using the EMBOSS dotmatcher tool, with different values of **word size** and **threshold**. Fig a reflects window size=7, threshold=20. Fig b shows window size=10, threshold=30. Fig c depicts windowlike=20, threshold=50.*







Exercise 5

Figure 8 *BRCA1 and 53BP1 using the EMBOSS Needle tool, with different values of gap penalties . Fig a reflects gap penalty=1, extend penalty=0.5 Fig b showing gap penalty=20, extend penalty=1. Fig c depicts gap penalty=5, extend penalty=0.5*

EMBOSS explorer

OUTPUT FILE outfile

```
#######################################
# Program: needle
# Rundate: Mon  5 Jan 2026 17:53:41
# Commandline: needle
#    -auto
#    -asequence /var/lib/emboss-explorer/output/405926/.asequence
#    -bsequence /var/lib/emboss-explorer/output/405926/.bsequence
#    -gapopen 10
#    -gapextend 0.5
#    -noendweight
#    -brief
#    -outfile outfile
#    -aformat3 srspair
# Align_format: srspair
# Report_file: outfile
#######################################

#=======================================
#
# Aligned_sequences: 2
# 1: BRCA1_HUMAN
# 2: TP53B_HUMAN
# Matrix: EBLOSUM62
# Gap_penalty: 10.0
# Extend_penalty: 0.5
#
# Length: 2365
# Identity:     405/2365 (17.1%)
# Similarity:   702/2365 (29.7%)
# Gaps:         895/2365 (37.8%)
# Score: 357.0
#
```

EMBOSS explorer

OUTPUT FILE outfile

```
#######################################
# Program: needle
# Rundate: Mon  5 Jan 2026 17:55:00
# Commandline: needle
#    -auto
#    -asequence /var/lib/emboss-explorer/output/524327/.asequence
#    -bsequence /var/lib/emboss-explorer/output/524327/.bsequence
#    -gapopen 20
#    -gapextend 1
#    -noendweight
#    -brief
#    -outfile outfile
#    -aformat3 srspair
# Align_format: srspair
# Report_file: outfile
#######################################

#=======================================
#
# Aligned_sequences: 2
# 1: BRCA1_HUMAN
# 2: TP53B_HUMAN
# Matrix: EBLOSUM62
# Gap_penalty: 20.0
# Extend_penalty: 1.0
#
# Length: 3789
# Identity:      10/3789 ( 0.3%)
# Similarity:    19/3789 ( 0.5%)
# Gaps:        3743/3789 (98.8%)
# Score: 9.0
#
#
```

EMBOSS explorer

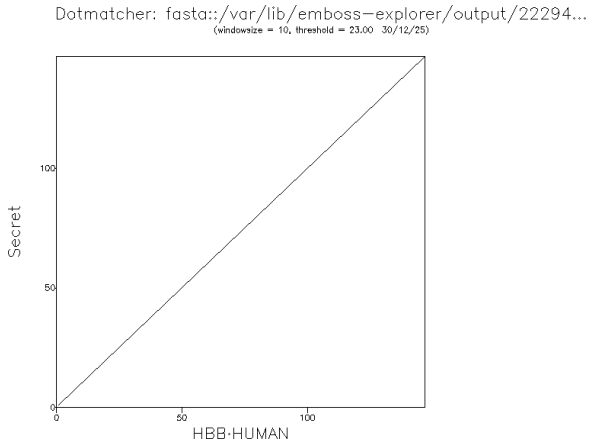
OUTPUT FILE outfile

```
#######################################
# Program: needle
# Rundate: Mon  5 Jan 2026 17:56:25
# Commandline: needle
#    -auto
#    -asequence /var/lib/emboss-explorer/output/597868/.asequence
#    -bsequence /var/lib/emboss-explorer/output/597868/.bsequence
#    -gapopen 5
#    -gapextend 0.5
#    -noendweight
#    -brief
#    -outfile outfile
#    -aformat3 srspair
# Align_format: srspair
# Report_file: outfile
#######################################

#=======================================
#
# Aligned_sequences: 2
# 1: BRCA1_HUMAN
# 2: TP53B_HUMAN
# Matrix: EBLOSUM62
# Gap_penalty: 5.0
# Extend_penalty: 0.5
#
# Length: 2411
# Identity:     563/2411 (23.4%)
# Similarity:   870/2411 (36.1%)
# Gaps:         987/2411 (40.9%)
# Score: 1195.0
#
#
```

Exercise 6

Figure S Needle Alignment of HBA (left) and HBB(right) with their respective secret sequences provided. A windowsize of 10 and threshold of 23. We can see an almost perfect alignment in both cases



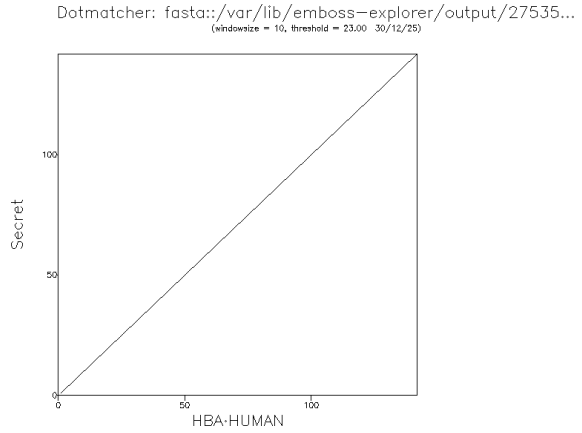
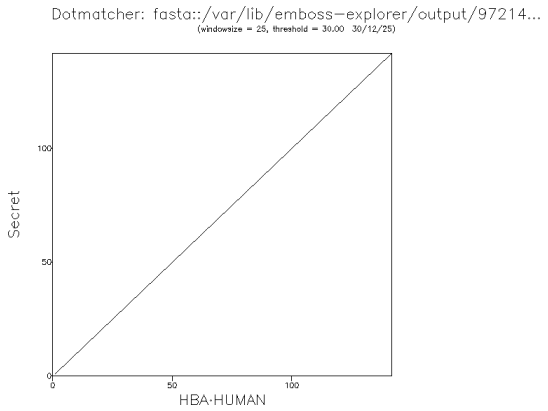


Figure 10 Needle Alignment of HBA (left) and HBB(right) with their respective secret sequences provided. A window size of 25 and threshold of 30. We can see a perfect alignment still in HBA but we interestingly observe a disturbance with HBB



Exercise 7

Figure 11Needle alignment of HBA(up) and HBB(down) with their corresponding secret sequences. We observed a perfect alignment for HBA at 100% and a minor difference with HBB at about SS.3%.

```
#    -bsequence /var/lib/emboss-explorer/output/054759/.bsequence
#    -gapopen 10
#    -gapextend 0.5
#    -noendweight
#    -brief
#    -outfile outfile
#    -aformat3 srspair
# Align_format: srspair
# Report_file: outfile
#######################################

#=======================================
#
# Aligned_sequences: 2
# 1: HBA_HUMAN
# 2: Secret
# Matrix: EBLOSUM62
# Gap_penalty: 10.0
# Extend_penalty: 0.5
#
# Length: 142
# Identity:     142/142 (100.0%)
# Similarity:   142/142 (100.0%)
# Gaps:           0/142 ( 0.0%)
# Score: 733.0
#
#
#=======================================

HBA_HUMAN          1 MVLSPADKTNVKAAWGKVGAHAGEYGAEALERMFLSFPTTKTYFPHFDLS     50
                     ||||||||||||||||||||||||||||||||||||||||||||||||||
Secret             1 MVLSPADKTNVKAAWGKVGAHAGEYGAEALERMFLSFPTTKTYFPHFDLS     50

HBA_HUMAN         51 HGSAQVKGHGKKVADALTNAVAHVDDMPNALSALSDLHAHKLRVDPVNFK    100
                     ||||||||||||||||||||||||||||||||||||||||||||||||||
Secret            51 HGSAQVKGHGKKVADALTNAVAHVDDMPNALSALSDLHAHKLRVDPVNFK    100

HBA_HUMAN        101 LLSHCLLVTLAAHLPAEFTPAVHASLDKFLASVSTVLTSKYR    142
                     ||||||||||||||||||||||||||||||||||||||||||
Secret           101 LLSHCLLVTLAAHLPAEFTPAVHASLDKFLASVSTVLTSKYR    142
```

```
#    -gapopen 10
#    -gapextend 0.5
#    -noendweight
#    -brief
#    -outfile outfile
#    -aformat3 srspair
# Align_format: srspair
# Report_file: outfile
#######################################

#=======================================
#
# Aligned_sequences: 2
# 1: HBB_HUMAN
# 2: Secret
# Matrix: EBLOSUM62
# Gap_penalty: 10.0
# Extend_penalty: 0.5
#
# Length: 147
# Identity:     146/147 (99.3%)
# Similarity:   146/147 (99.3%)
# Gaps:           0/147 ( 0.0%)
# Score: 773.0
#
#
#=======================================

HBB_HUMAN          1 MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLS     50
                     ||||||.||||||||||||||||||||||||||||||||||||||||||
Secret             1 MVHLTPVEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLS     50

HBB_HUMAN         51 TPDAVMGNPKVKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLHVD    100
                     ||||||||||||||||||||||||||||||||||||||||||||||||||
Secret            51 TPDAVMGNPKVKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLHVD    100

HBB_HUMAN        101 PENFRLLGNVLVCVLAHHFGKEFTPPVQAAYQKVVAGVANALAHKYH    147
                     |||||||||||||||||||||||||||||||||||||||||||||||
Secret           101 PENFRLLGNVLVCVLAHHFGKEFTPPVQAAYQKVVAGVANALAHKYH    147


#---------------------------------------
#---------------------------------------
```

Exercise 10

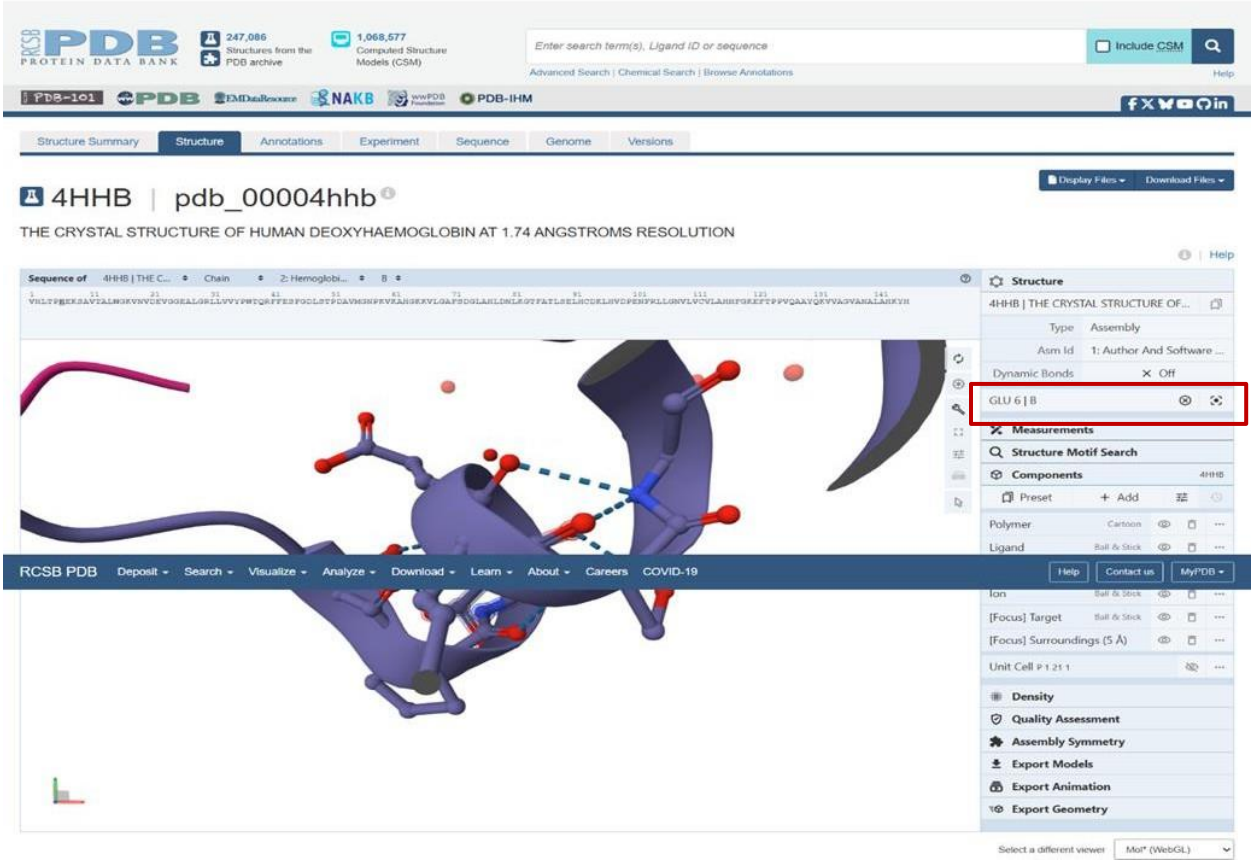


Figure 12 Inspection of ***4HHB and*** *specifically examining* ***residue 6 of the β chain****, we observed that In* ***4HHB (normal hemoglobin B)****, residue* ***β6 is Glutamate (Glu, E)***

Figure 13 Analysis of Hemoglobin S on PDB (2HBS) , and further examination of the residue β6 reveals Valine (Val, V) instead of Glutamate, E.